AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 23 · STUDENT LESSON

Block to Python Bridge

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Before you start

AI-2 Weeks 4–5: trace assignment and loops before translating syntax.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Translate a short paper calculation into a Python script. Try the recall before opening notes.

Your learning target

I can translate a short paper calculation into a Python script.

I can predict and explain the order of executed statements.

I can keep expected and observed **output** separate when checking a **program**.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Requests Responses and JSON. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Conversation State and Memory. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Model Memory vs Retrieval. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 23: Block-to-Python Bridge

Variables, **condition**s, **list**s, loops, and **function**s carry the same logic across representations.



CREATOR MISSION

Translate a block workflow into readable Python.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

> Four packs contain six counters each. What steps find and show the total?

This week’s end goal

By the end, I can explain block to python bridge using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Block to Python Bridge case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use block-to-python bridge to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

In Grade 6 you traced rules on paper. Now you will write a small Python **program** and check what it actually does. Begin with a counter-packing calculation, not the whole activity adviser.

The requirement is to find the total counters in equal packs and display that total. For this first **program**, the supplied quantities are whole, nonnegative numbers; it does not yet validate typed **input**.

Use a teacher-prepared Python 3 editor that can run a saved .py file. Open a new file called counters.py, type the code exactly as shown in the next section, save it, and use the editor's Run command.

Read the **output** area separately from the source editor. No account, AI tool, package installation or personal data is needed. If a device is unavailable, trace on paper and label the result predicted rather than executed.

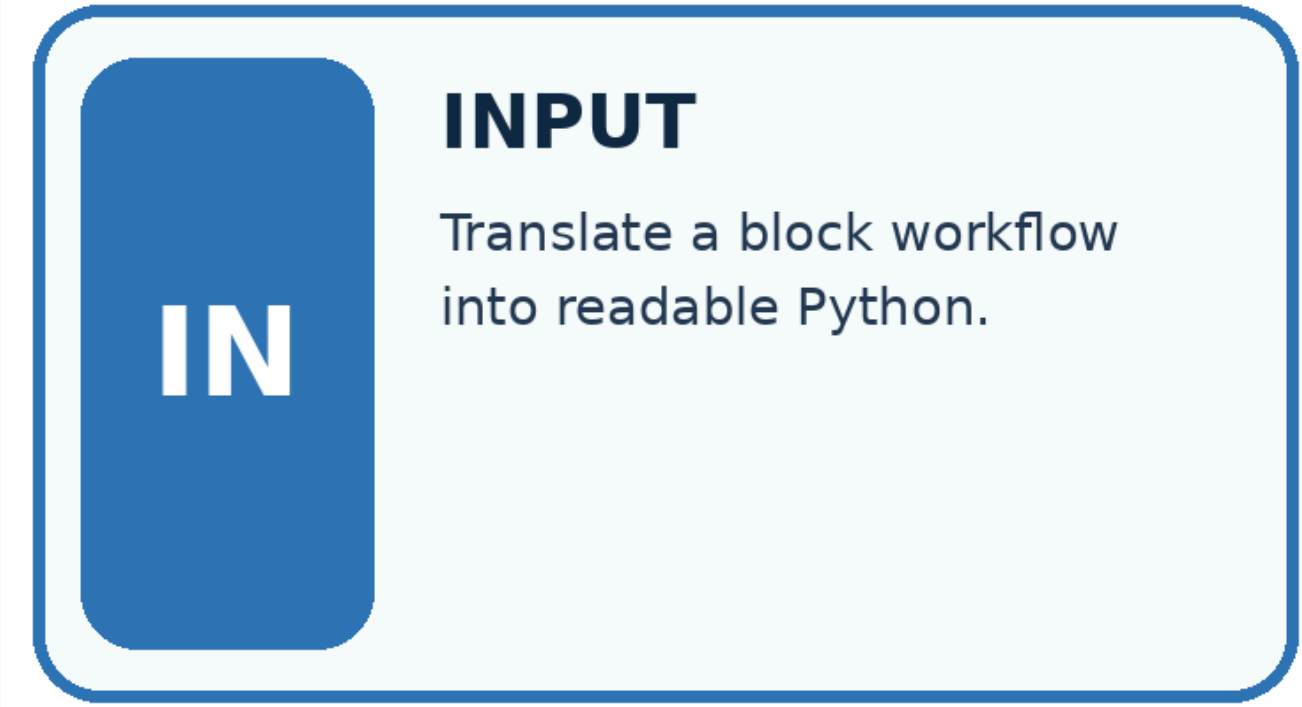
See how it works

> Follow the mechanism for Block to Python Bridge. Connect each labelled input, deciding step and result.

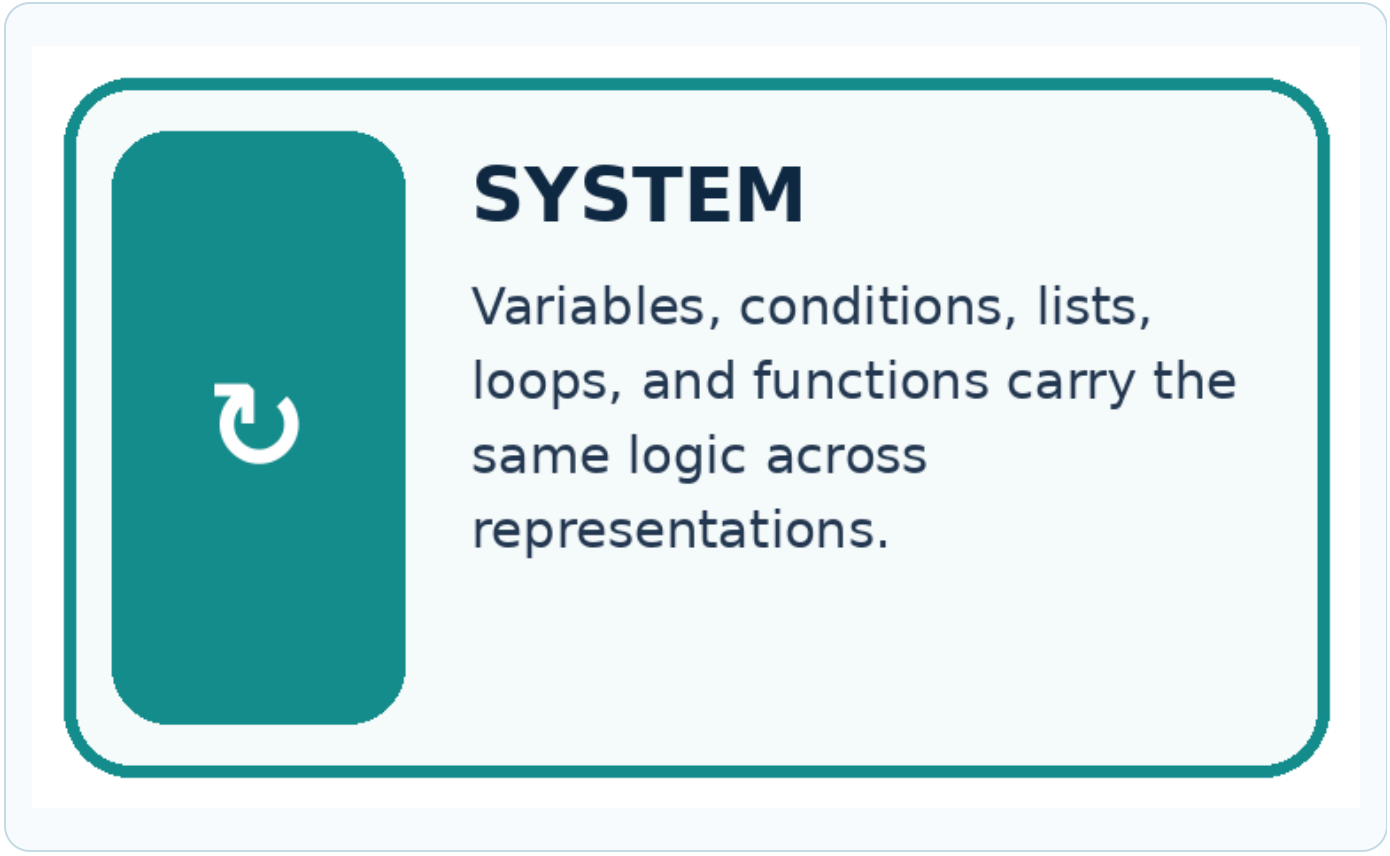
VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓


SYSTEM
Variables, conditions, lists, loops, and functions carry the same logic across representations.

↓


OUT
EVIDENCE
Translate a block workflow into readable Python.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

A calculation from paper to output

PAPER STEP	PYTHON INSTRUCTION	EFFECT
Record pack count	packs = 4	Name holds 4
Multiply two quantities	total = packs * per_pack	Name holds product
Show the answer	print(total)	Displays the value

Assign per_pack before the multiplication. This diagram summarises the steps; the complete script below includes that assignment.

```
packs = 4
per_pack = 6
total = packs * per_pack
print(total)
```

Read the script from top to bottom. The equals sign assigns the value on its right to the name on its left. The star means multiplication. print(total) displays the value of total; it does not display the letters in the name.

Type ordinary straight characters and keep the parentheses. The Python prompt symbols >>> are not part of this saved script.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

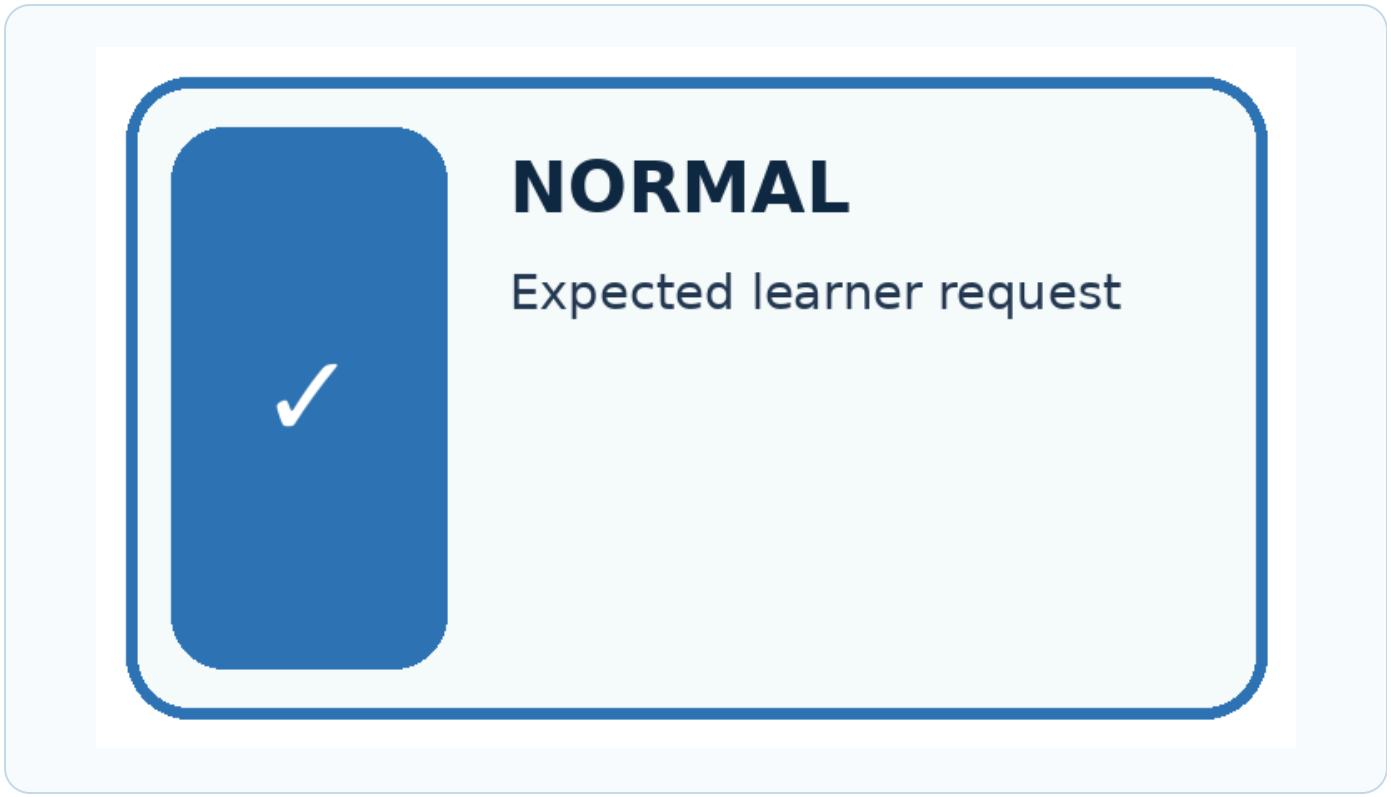
> Compare the supplied cases for Block to Python Bridge. Identify the detail that changes the conclusion.

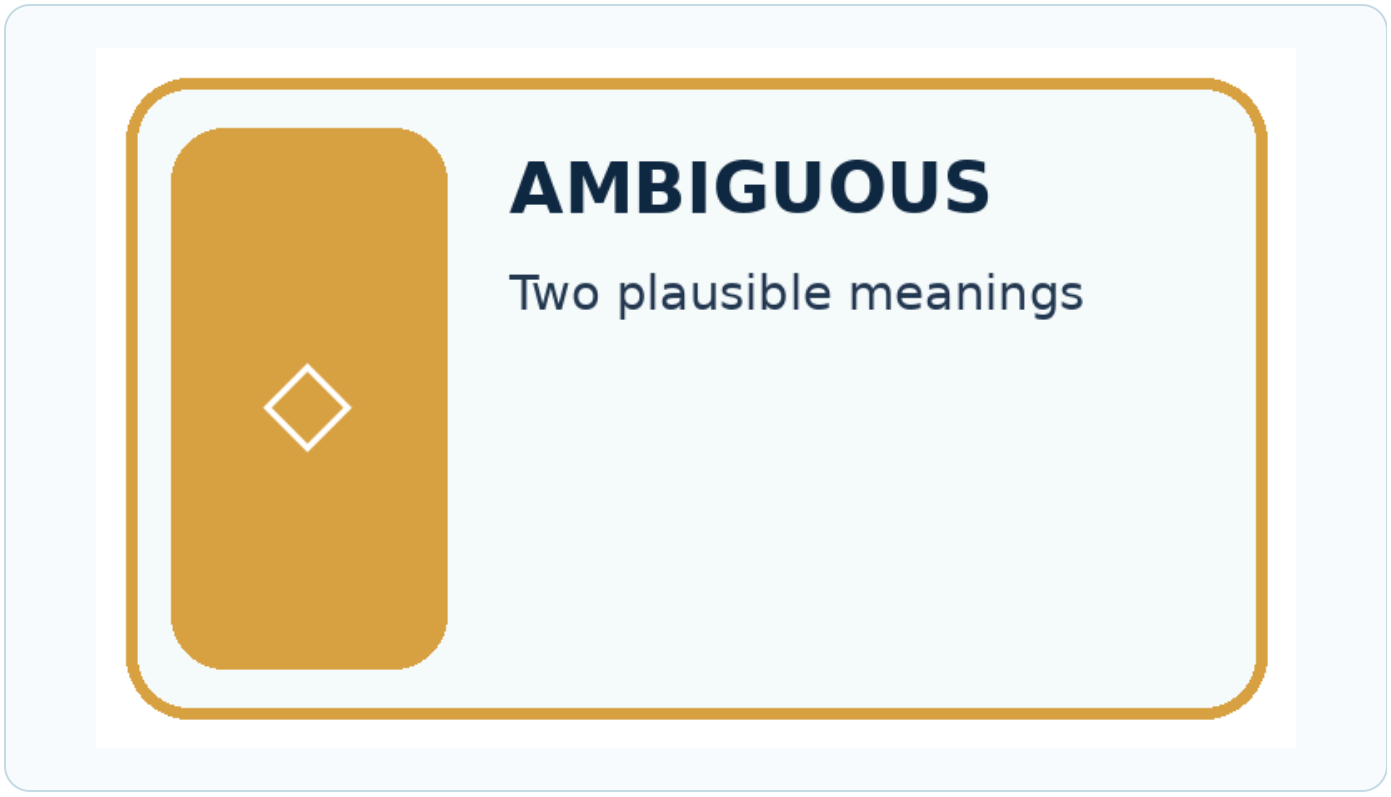
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.


NORMAL
Expected learner request


AMBIGUOUS
Two plausible meanings

The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Replace the last page count 90 with 100.

Predicted result

Prediction: The final count becomes 1.

Why the result follows

Reason: The comparison is strictly less than 100, so the boundary value does not qualify.

STUDENT LESSON · 5 OF 18

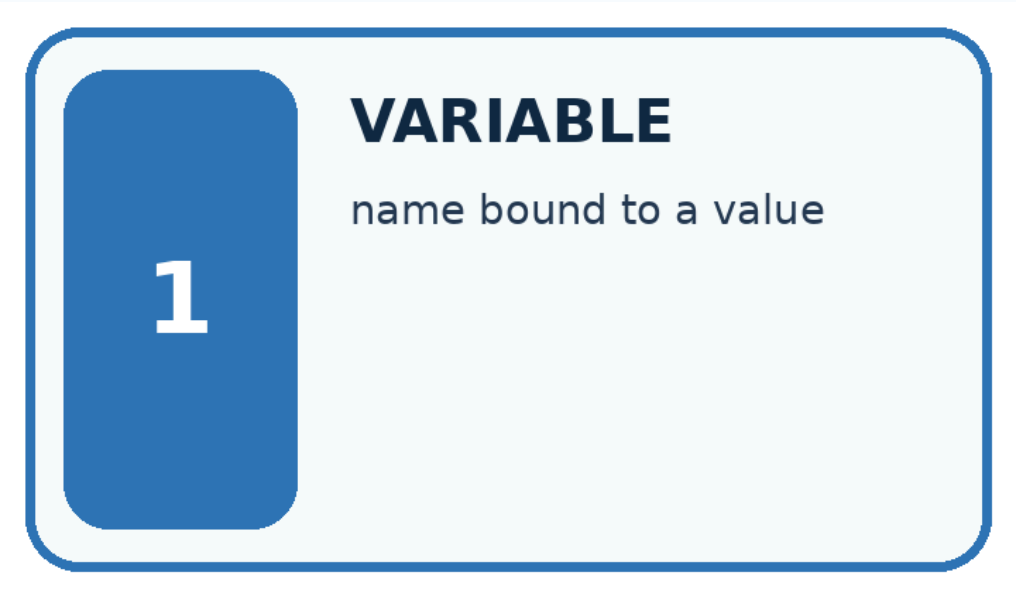
Words that unlock the idea

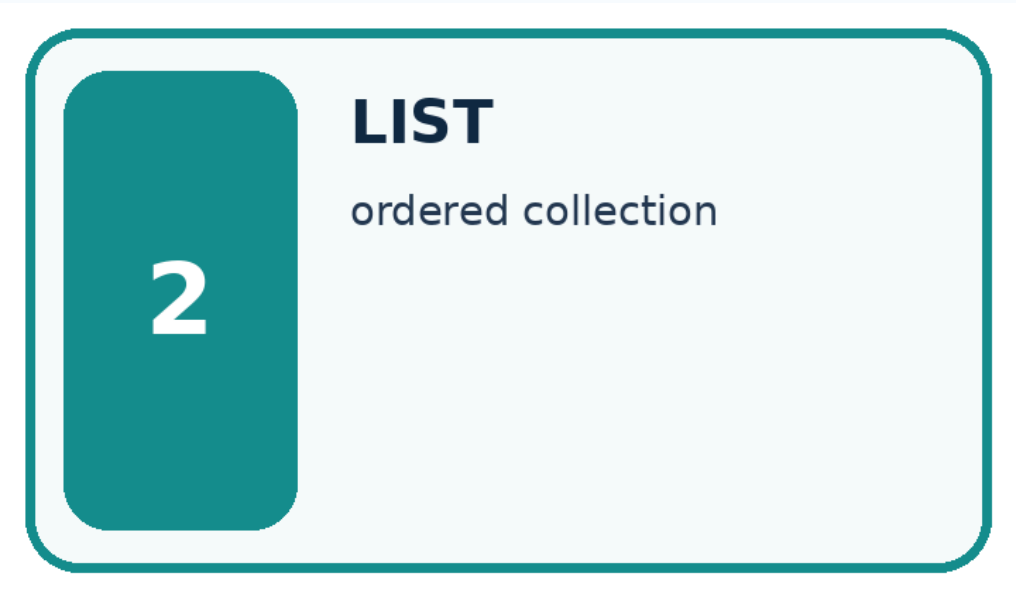
> Use the vocabulary of Block to Python Bridge to explain a concrete decision, rather than only memorising definitions.

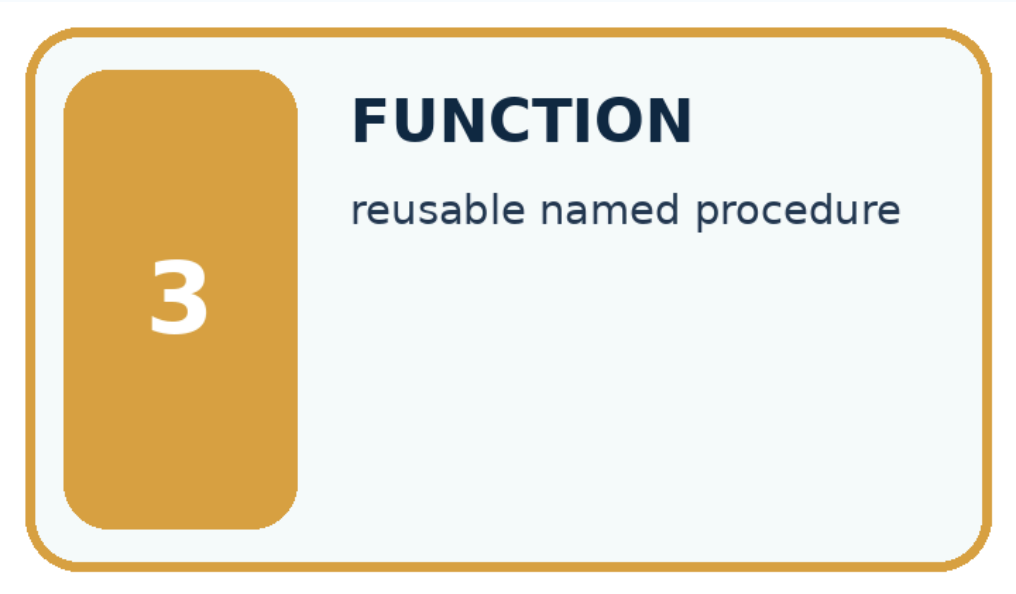
VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.


1
VARIABLE
name bound to a value


2
LIST
ordered collection





Word	My own example	Not this / boundary
variable	A named reference to a value that can be used or changed in a program.	
list	An ordered collection of items.	
function	A named unit of code that performs a task, optionally receiving arguments and returning a value.	

Word	My own example	Not this / boundary
condition	A statement evaluated as true or false to help select an action.	

Words used in the closer explanation

Word	Meaning
Algorithm	An ordered procedure for solving a stated problem
Program	Instructions written in a language a computer can execute
Interpreter	Software that executes Python instructions
Assignment	Binding a name to the value of an expression
Output	Information a program displays or returns

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Block to Python Bridge, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 23 • WORKED EVIDENCE

1. Block: for each passage, check relevance.
2. Python: for passage in passages: ...
3. Condition keeps passages with score ≥ threshold.
4. A function packages the selection logic and can be tested with normal and edge lists.

1. Block: for each passage, check relevance.

2. Python: for passage in passages: ...

3. **Condition** keeps passages with score ≥ threshold.

4. A **function** packages the selection logic and can be tested with normal and edge **list**s.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

After line	Names and values	Displayed output
packs = 4	packs is 4	Nothing yet
per_pack = 6	packs is 4; per_pack is 6	Nothing yet
total = packs * per_pack	total is 24	Nothing yet
print(total)	Values are unchanged	24

Worked explanation · Block to Python Bridge

Write 24 as the expected **output** before running. Then copy the actual **output** into a separate column. A matching result supports this case, not every possible **input**.

A saved file may contain an old edit: save again and rerun the whole script. Do not type the expected number into the **output** record without running.

Why is there no displayed output after the third line? What changes if print(total) is placed first in a fresh run, before any assignment? Explain using which names exist.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Block to Python Bridge

Translate a “count qualifying books” paper procedure into a Python-style trace.

The facts we will use

Supplied facts

Page counts are [80, 150, 90]. Start count=0; for each pages value, add 1 only when pages<100.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/23

Block to Python Bridge

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Start counter

count begins at 0 before the loop.



2. Trace decisions

80 qualifies: count 1. 150 does not: count 1. 90 qualifies: count 2.



3. Report count

The final result is 2 qualifying books, not the sum of their page counts.

CHANGE ONE THING

Replace the last page count 90 with 100.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Start counter	count begins at 0 before the loop.
2. Trace decisions	80 qualifies: count 1. 150 does not: count 1. 90 qualifies: count 2.
3. Report count	The final result is 2 qualifying books, not the sum of their page counts.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. The final result is 2 qualifying books, not the sum of their page counts. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

This deterministic loop counts a stated **condition**; it neither trains a model nor measures whether the books suit a reader.

Keep the stated limit

Read the labelled picture: Trace the changed **list** and identify where the result differs. Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Replace the last page count 90 with 100. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Block to Python Bridge claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Moving blocks into text automatically creates correct Python. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

Save the file with the **input** values, expected and observed **output**s, and a short explanation. This **program** follows a written multiplication rule.

It has not trained a model, predicted somebody's preferences or checked that real packs contain the stated number. Its conclusion depends on the supplied data. Week 2 examines the values and types behind these names.

A script runs without an error and displays 11 for five packs of six. Is it correct for the stated requirement? Give the required result, the likely wrong operation and one check after repair. State whether your evidence came from a device run or a paper trace.

STUDENT LESSON · 8 OF 18

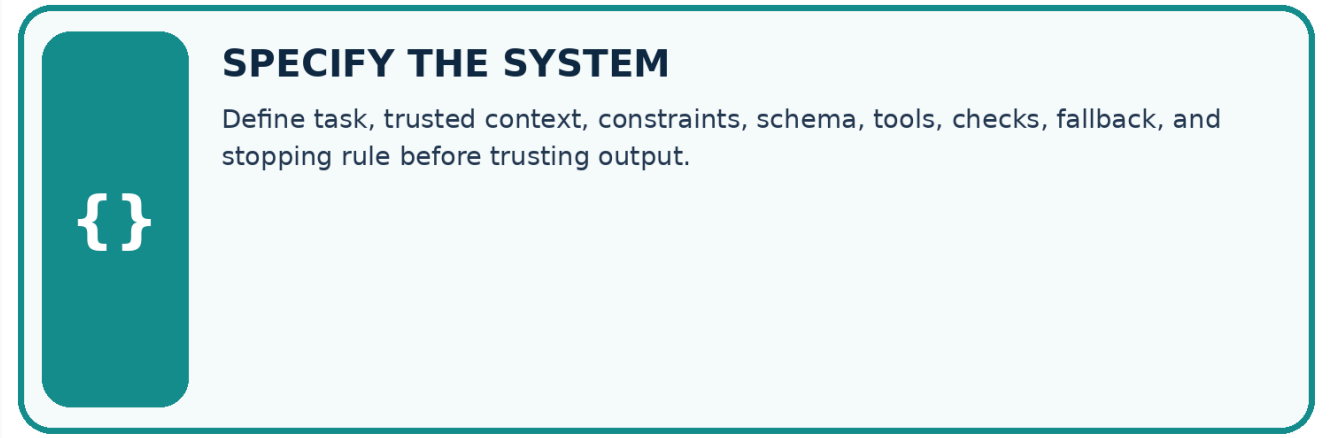
Practise together

Practise Block to Python Bridge with support, then explain the deciding step in your own words.

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Change only packs to 5. Predict the result from five groups of six, save, then run the entire file again. The expected **output** is 30. Keep the original four-pack result beside the new one and label the changed **input**.

A comment can explain an instruction: Python ignores text after # outside a string. A comment does not change the calculation.

```
packs = 5  # five equal packs
per_pack = 6
```

```
total = packs * per_pack
print(total)
```

A **program** can run successfully and still solve the wrong problem. Replacing * with + would display 11 for this **input**, but addition does not count five groups of six.

Use the requirement and a paper calculation to check meaning; the absence of an error message is not proof of correctness.

Practise with less help: Predict then observe

Use workbook W2 for the connected example “From a paper **algorithm** to Python”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for From a paper **algorithm** to Python. Proposed explanation: “An equals sign displays the answer”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Trace the changed **list** and identify where the result differs. Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Block to Python Bridge. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Translate five blocks, predict **variable** values, run or simulate, then test empty-**list** and missing-field cases.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** missing
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** fresh transfer
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

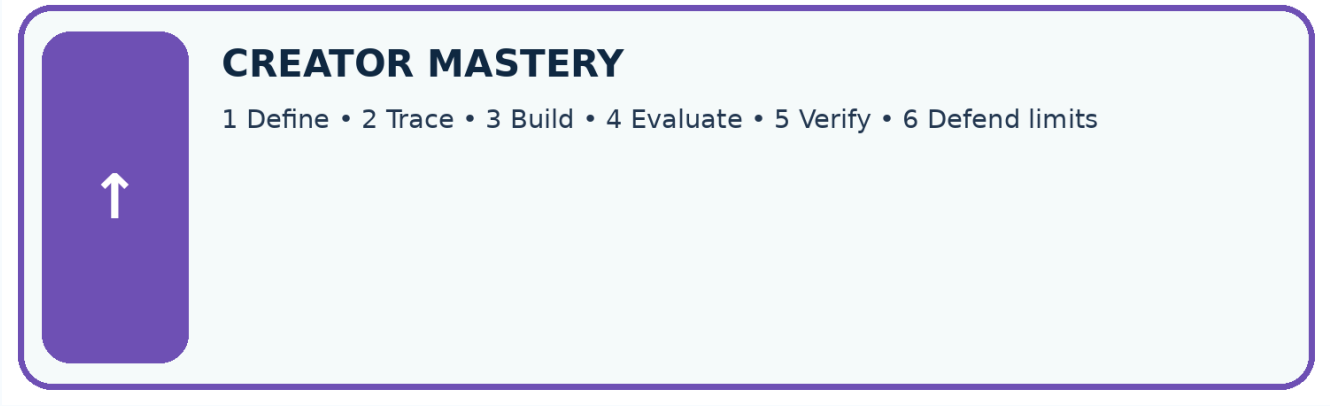
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Block to Python Bridge. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

> **Write or draw your reasoning and evidence.**

STUDENT LESSON · 11 OF 18

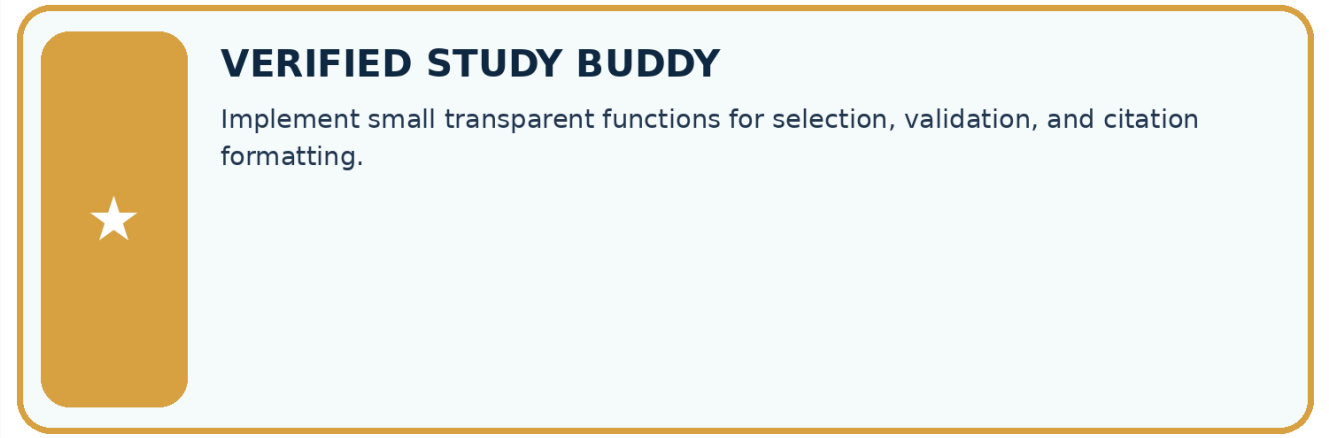
Connect your project

> Apply this week's idea to Verified Study Buddy. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Implement small transparent **function**s for selection, validation, and citation formatting.

Trace the system

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent **output** with verified truth.

Evidence for your project

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

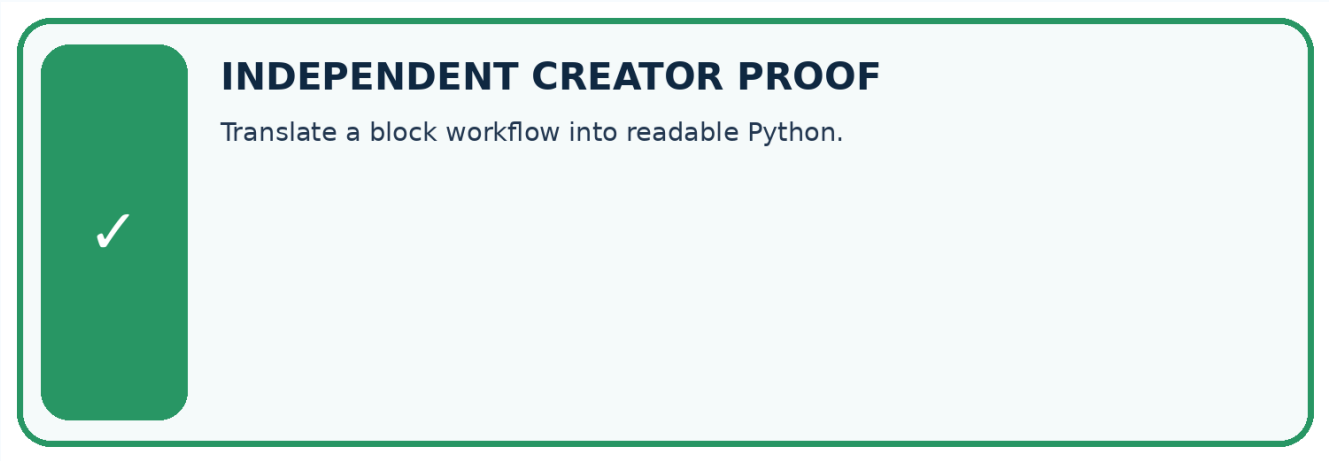
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Translate a block workflow into readable Python.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Block to Python Bridge case for supported practice. Repeated examples do not become fresh assessments.

Week 23 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A block **program** sets total=0 and adds each value in [2,4,6].

The question

Translate and trace.

Worked explanation

Python uses total=0, then an indented total += value inside for value in [2,4,6]. Totals are 2,6,12.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Trace the accumulator inside the loop

Evidence or stage	Value or result
Start total	0
Add 2	2
Add 4	6

Evidence or stage	Value or result
Add 6	12

A block **program** sets total=0 and adds each value in [2,4,6].

Translate and trace.

Python uses total=0, then an indented total += value inside for value in [2,4,6]. Totals are 2,6,12.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Changed independent case

Set total=0, then for values2,4,9 perform total=total+value. Trace the result. What changes if total=0 moves inside the loop before that addition?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 23 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> Trace this Python: score = 2; then if score >= 2: score = score + 3. What is the final value, and what changes if the **condition** becomes score > 2?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Implement small transparent **function**s for selection, validation, and citation formatting.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Write Python that totals [3,5,7], then trace each total. Explain how indenting 'total=0' inside the loop changes the result.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

Use this map to navigate Block to Python Bridge, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Translate a “count qualifying books” paper procedure into a Python-style trace.

By the end, I can	How I will show it
Translate a short paper calculation into a Python script	Explain the deciding step, show a trace or test, and state one limit.
Predict and explain the order of executed statements	Explain the deciding step, show a trace or test, and state one limit.
Keep expected and observed output separate when checking a program	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Block to Python Bridge without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Translate a short paper calculation into a Python script

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Translate a “count qualifying books” paper procedure into a Python-style trace.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The final result is 2 qualifying books, not the sum of their page counts.

1. Trace decisions

2. Start counter

3. Report count

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Report count The final result is 2 qualifying books, not the sum of their page counts. Read the picture in order.

The final result is 2 qualifying books, not the sum of their page counts. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Replace the last page count 90 with 100. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. The final count becomes 1.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed **condition** and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The final count becomes 1. The comparison is strictly less than 100, so the boundary value does not qualify.

Repair a missing building block

Review Requests Responses and JSON (ai-3/22). Goal: Parse JSON into Python records. Short practice: Would converting the text silently prove the source count is correct?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Conversation State and Memory (ai-3/20). Goal: Explain the **mechanism** and terminology of conversation state and retention. Short practice: What is selected_book after the correction and after exit?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Model Memory vs Retrieval (ai-3/17). Goal: Explain the **mechanism** and terminology of retrieval and indexing. Short practice: Why can the assistant not certify 15: 00 when the notice is missing?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

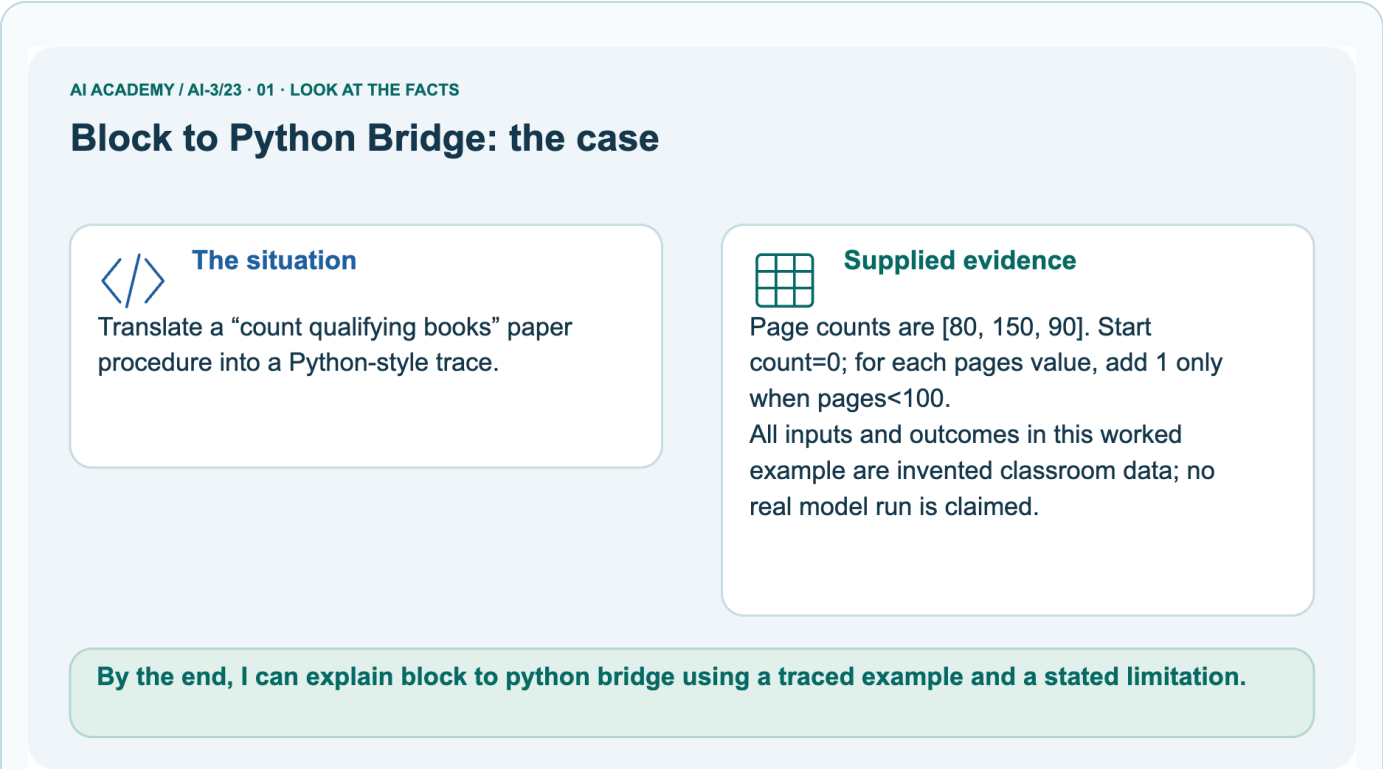
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Block to Python Bridge graphic using its labels, deciding evidence and relevant limitation.

01. Block to Python Bridge: the case



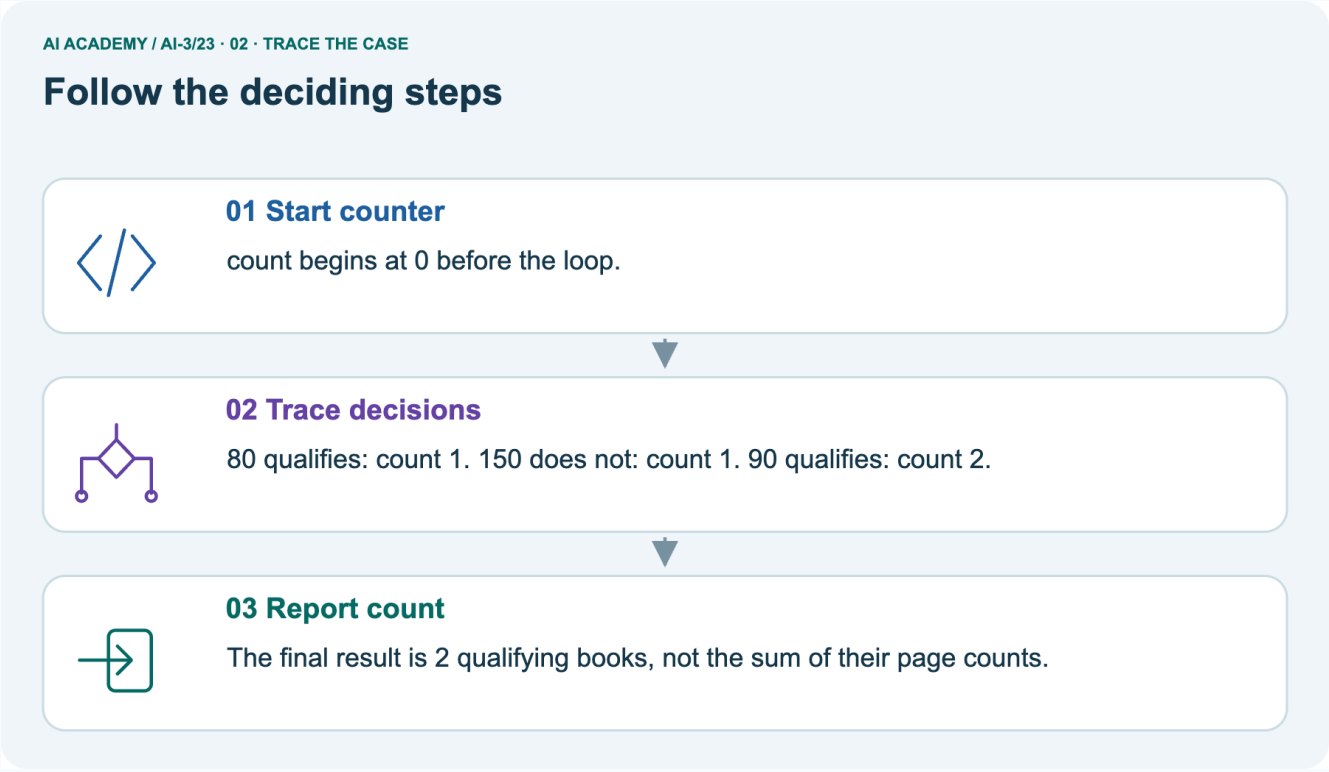
Follow the labelled picture with the matching explanation. Open the vector diagram

Translate a “count qualifying books” paper procedure into a Python-style trace. Supplied facts: Page counts are [80, 150, 90]. Start count=0; for each pages value, add 1 only when pages<100.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Start counter: count begins at 0 before the loop. → Trace decisions: 80 qualifies: count 1. 150 does not: count 1. 90 qualifies: count 2. → Report count: The final result is 2 qualifying books, not the sum of their page counts.

> Point to each numbered stage and explain which recorded **input** it uses.

03. A calculation from paper to output

AI ACADEMY / AI-3/23 · 03 · INSPECT THE LESSON MODEL

A calculation from paper to output

Paper step	Python instruction	Effect
Record pack count	packs = 4	Name holds 4
Multiply two quantities	total = packs * per_pack	Name holds product
Show the answer	print(total)	Displays the value

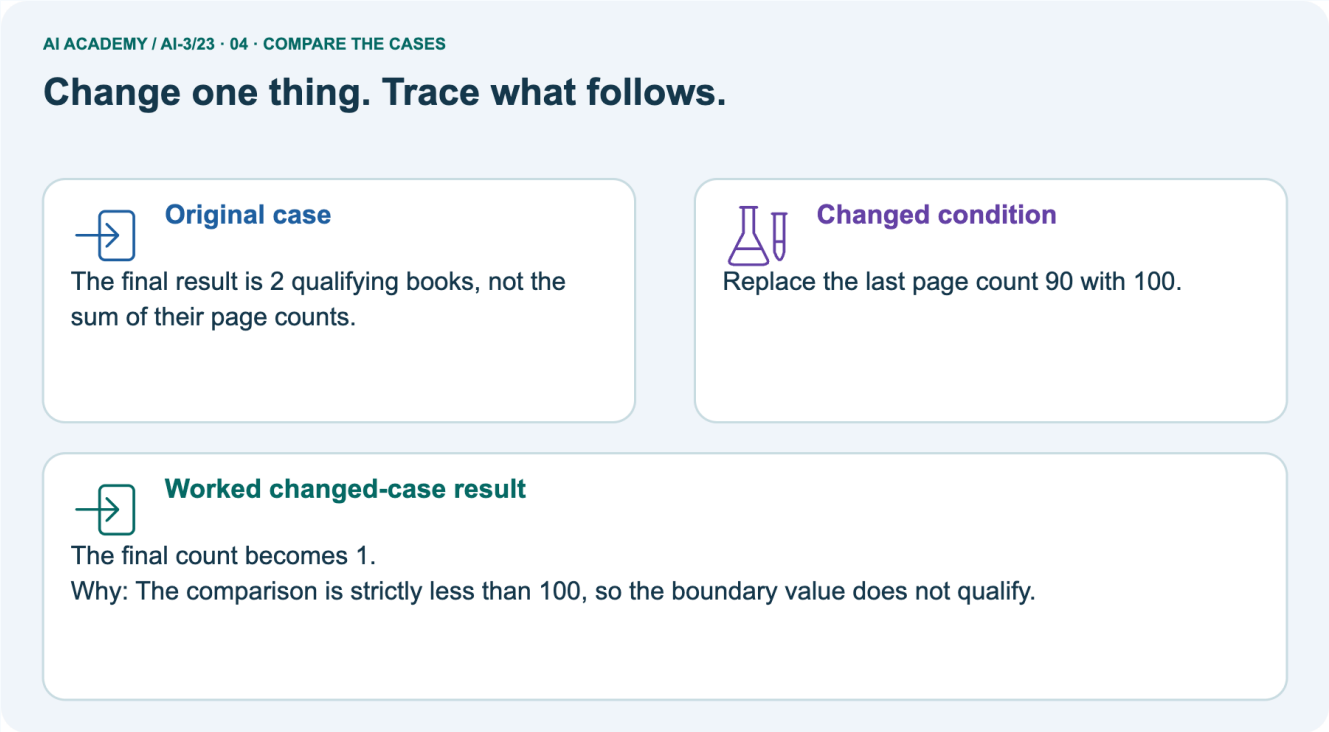
Assign per_pack before the multiplication. This diagram summarises the steps; the complete script below includes that assignment.

Follow the labelled picture with the matching explanation. Open the vector diagram

A calculation from paper to **output**. Record pack count | packs = 4 | Name holds 4. Multiply two quantities | total = packs * per_pack | Name holds product. Show the answer | print(total) | Displays the value

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



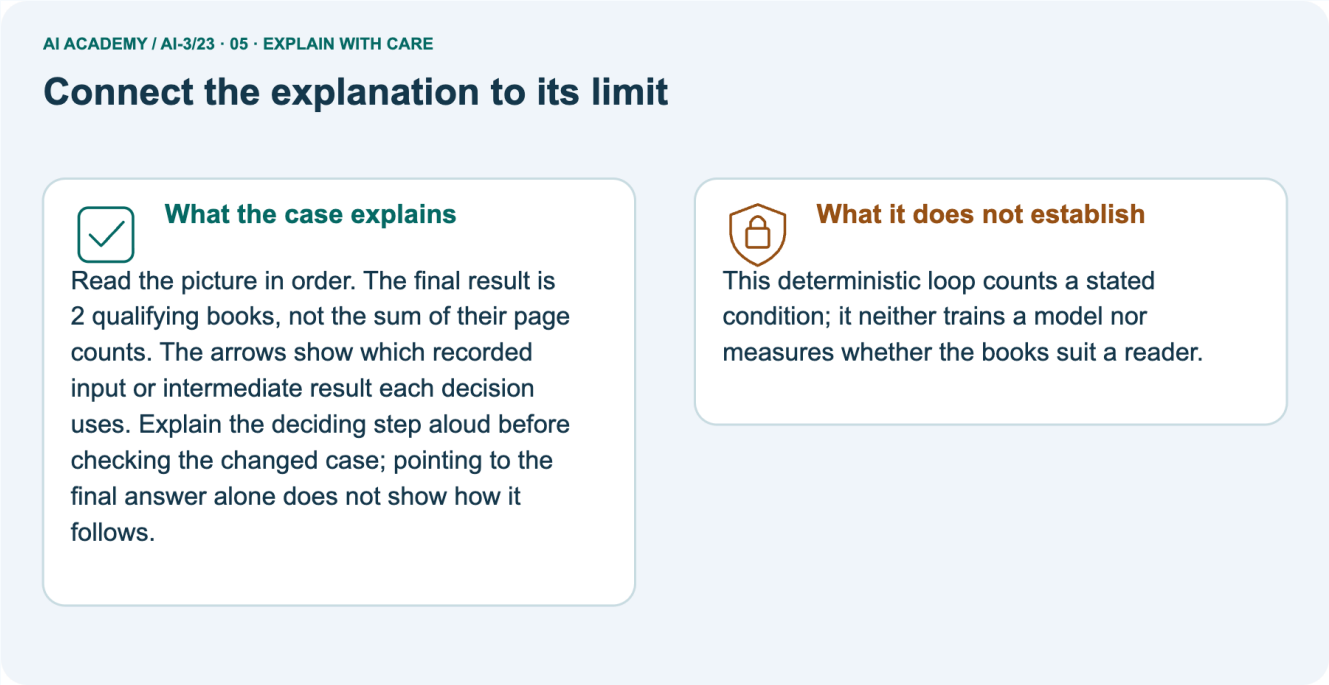
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The final result is 2 qualifying books, not the sum of their page counts. Changed **condition**: Replace the last page count 90 with 100. Expected result: The final count becomes 1.

Reason: The comparison is strictly less than 100, so the boundary value does not qualify.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

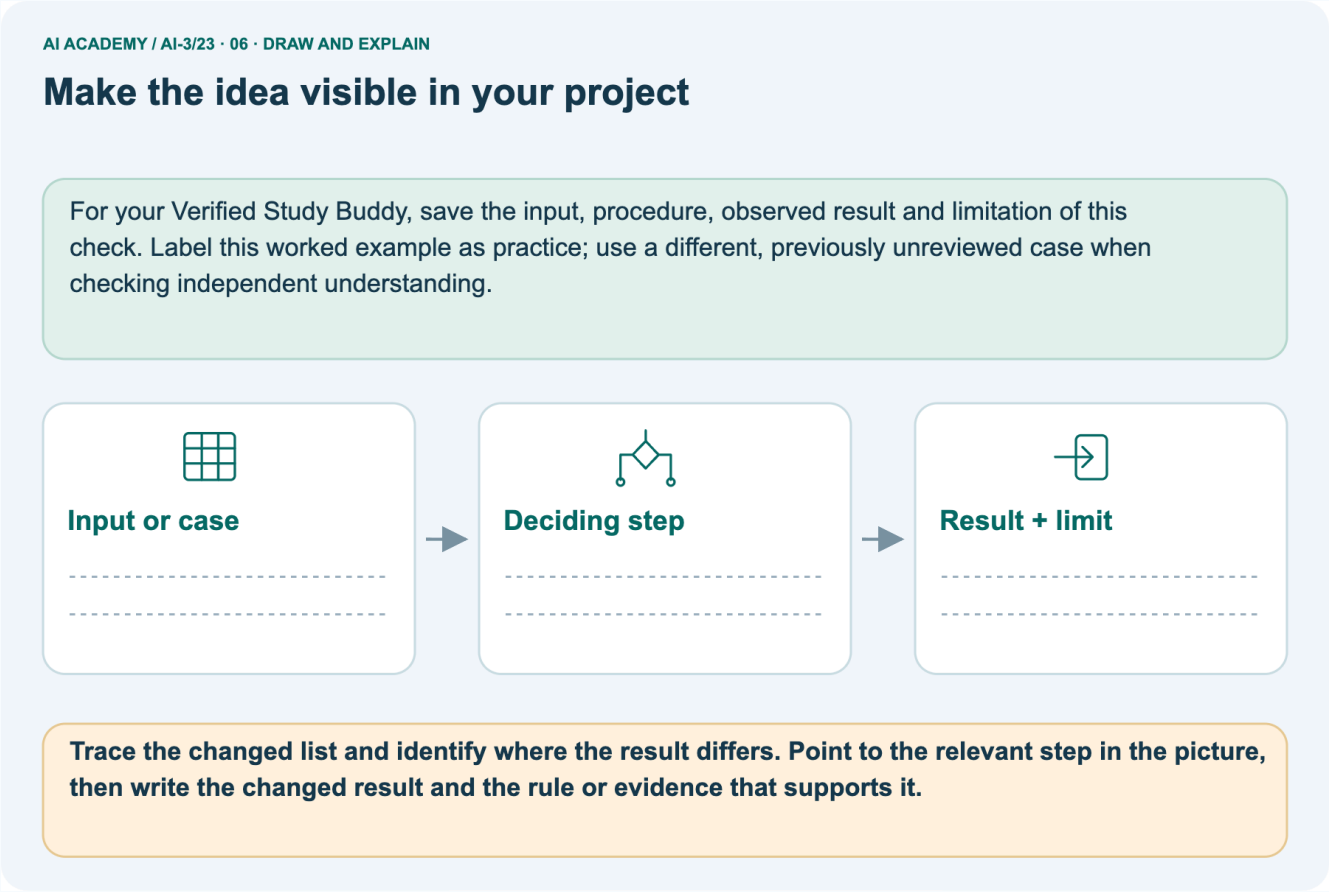
Read the picture in order. The final result is 2 qualifying books, not the sum of their page counts. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: This deterministic loop counts a stated **condition**; it neither trains a model nor measures whether the books suit a reader.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Trace the changed **list** and identify where the result differs. Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.